

Centrofacial Lentiginosis

Lentigines, which are dark brown to gray-blue, appear on the lips, around the mouth, and on the fingers. Lip macules may, over time, disappear. Macules of the buccal mucosa are blue to blue-black and are pathognomonic; unlike lip lesions, these do not tend to disappear with time. Characteristic lentigines usually develop during childhood and in the first months of life. Clinical diagnosis is primarily based on the typical facial features and the presence of hypertrophic cardiomyopathy and/or café-au-lait macules. The disorder is caused by mutations in the *PTPN11* gene, coding for the protein tyrosine phosphatase SHP-2, and situated on chromosome 12. LEOPARD syndrome (LS) is allelic to Noonan syndrome and shares several clinical features.

Congenital Hyperpigmentation Lait Macules	Circumscribed with Café-au-
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Café-au-lait macules (CALMs) consist of sharply and irregularly bordered hyperpigmented patches of skin, varying in size from 0.5 cm to more than 20.0 cm. They are often present at birth or appear in the early months of life. Between 0.3 percent and 18 percent of all newborns display isolated CALMs. Histologically, isolated CALMs show a normal number of melanocytes but increased epidermal melanin (epidermal melanotic hypermelanosis). Multiple CALMs are well-known markers for several multisystem disorders.

Neurofibromatosis Type 1

Neurofibromatosis type 1 (NF1), first described by Friedrich von Recklinghausen in 1882, is also known as von Recklinghausen disease. It is an

autosomal dominant disorder caused by mutations in the *NF1* gene located on chromosome 17q11.2, which encodes the protein neurofibromin. Neurofibromin participates in multiple signaling pathways and undergoes alternative splicing to generate several isoforms. Its most critical function is the downregulation of the Ras signal transduction pathway, classifying it as a tumor suppressor gene.

NF1 is considered a neurocristopathy and is characterized by various cutaneous and systemic manifestations related to pigment cells, including café-au-lait macules (CALMs), intertriginous freckling, and iris Lisch nodules. The U.S. National Institutes of Health established diagnostic criteria in 1987, which include the presence of six or more CALMs—greater than 5 mm in diameter in prepubertal individuals or greater than 15 mm post-puberty—as a key diagnostic feature. Unlike isolated CALMs, NF1-associated CALMs show a significantly increased number of epidermal melanocytes. Intertriginous freckles, which are pathognomonic for NF1, also exhibit increased epidermal melanocytes, distinguishing them histologically from ephelides (simple freckles).

Watson Syndrome

Watson syndrome is a rare condition sharing clinical features with NF1, including pulmonary valve stenosis, CALMs, and short stature. It is associated with specific mutations in the *NF1* gene and may also include developmental delay and facial dysmorphism.

McCune-Albright Syndrome

McCune-Albright syndrome (MAS) was first described by McCune (1936) and Albright (1937) and classically presents as a triad of polyostotic or monostotic fibrous dysplasia, café-au-lait macules, and hyperfunctioning endocrinopathies such as precocious puberty, hyperthyroidism, hypercortisolism, hypersomatotropism, and hypophosphatemic rickets. The CALMs in MAS are typically fewer in number and have more irregular, "coast of Maine" borders compared to those in NF1. They are characteristically limited to one side of the body and respect the midline. MAS results from a post-zygotic activating mutation in the *GNAS* gene, encoding the α -subunit of the stimulatory G

protein (Gs α). This leads to mosaic activation of adenylate cyclase and constitutive cAMP signaling in affected tissues.

Nevus Spilus

Nevus spilus, also known as speckled lentiginous nevus, is a hyperpigmented macule or patch with smaller, darker macules or papules scattered across its surface. It may be present at birth or appear in early childhood. Histologically, the background patch shows features of a lentiginous melanocytic proliferation, while the darker speckles may resemble lentigines or nevi. Although generally benign, rare cases have been associated with malignant transformation, warranting clinical monitoring.